

Wi-Fi Series

Data Sheet

USE_8721

Wi-Fi+BLE Module

Version

VERSION	REVISION DATE	REVISOR	REVIEWER	REVISED CONTENTS
1.0	02/03/2026	MB	AD	Initial release

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Features:

General

- Dual-band Wi-Fi 802.11 a/b/g/n 1x1 (2.4 GHz & 5 GHz)
- Bluetooth 5.0 (LE-1M / LE-2M / LE-Coded PHY)
- Supports LE Data Length Extension
- Supports Link Layer Privacy
- Supports LE Advertising Extensions
- Supports Scatternet (8 Master + 3 Slave roles)
- Operating Temperature: -40°C to $+105^{\circ}\text{C}$
- Operating Voltage: $3.3\text{ V} \pm 10\%$

MCU Platform

- Chipset: Realtek RTL8721DAF-VT2-CG
- Dual-core Processor:
 - KM4 up to 345 MHz
 - KM0 up to 115 MHz
- On-chip SRAM:
 - 512 KB + 160 KB
- Embedded Flash: 4 MB (MCM)
- Secure boot supported

Wi-Fi Features

- IEEE 802.11 a/b/g/n
- Frequency Range:
 - 2.4 GHz: 2412 MHz – 2484 MHz
 - 5 GHz: 5180 MHz – 5825 MHz
- Channel bandwidth: 40 MHz
- Max data rate: MCS7
- Security:
 - WEP, WPA, WPA2, WPA3, WPS
- Authentication:
 - Open, shared key, pair-wise key
- Power saving mechanism
- RF coexistence between Wi-Fi and BLE

Bluetooth Low Energy

- Bluetooth 5.0 (LE-1M/LE-2M/LE-Coded PHY (long range))
- Supports piconets in a scatter-net (up to 8 master roles and 3 slave roles)

- Co-existence RF design between Wi-Fi and Bluetooth

Peripheral Interfaces

- 4x UART interface, baud rate up to 8Mbps within high-speed mode (40MHz)
- 2 x I2C, three speed modes: standard up to 100Kbps, fast up to 400Kbps, high to 3.4Mbps
- SDIO Device, Default speed mode (25MHz) and high-speed mode (50MHz)
- 2 x SPI, up to 50Mbps in master and slave modes, in half-duplex, full-duplex and simplex modes.
- 8 x PWM with configurable duration and duty cycle from 0 ~ 100%
- Cap touch x 4 channels
- ADC x (7+1) channels
- Programmable GPIOs
- USB 2.0 Device (Full Speed)
- IR

General Description:

The **USE_8721** is a high-performance multi-radio MCU module designed for advanced wireless applications. Built on a dual-core architecture, it enables customers to develop sophisticated applications running on two RISC cores. The module supports dual-band Wi-Fi IEEE 802.11a/b/g/n (up to 40 MHz bandwidth) and Bluetooth Low Energy (BLE) 5.0 connectivity.

With its rich set of peripherals and powerful processing capability, the USE_8721 is an ideal solution for smart home systems, industrial automation, IoT gateways, and consumer electronics.

For more information, reach out to us at info@unisemindia.com.

Pinout Diagram:

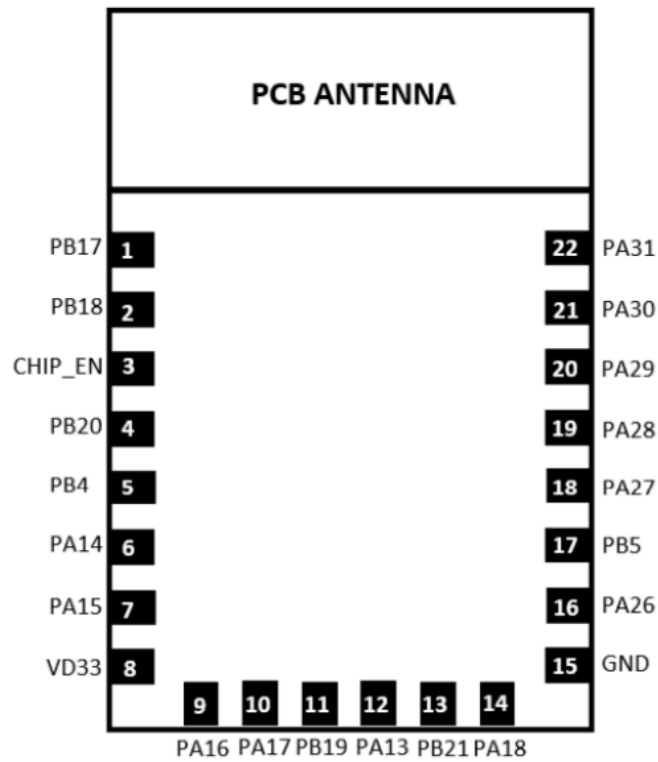


Fig1: Module Pinouts

Pin Descriptions:

Pin No.	Pin Name	Type	Function Description
1	PB17	I/O	GPIOB_17
2	PB18	I/O	GPIOB_18
3	CHIP_EN	I	High: Enable the chip Low: Module power off
4	PB20	I/O	GPIOB_20
5	PB4	I/O	GPIOB_4 / UART_LOG_RXD
6	PA14	I/O	GPIOA_14

7	PA15	I/O	GPIOA_15
8	VD33	P	3.3 V Power Supply
9	PA16	I/O	GPIOA_16
10	PA17	I/O	GPIOA_17
11	PB19	I/O	GPIOB_19
12	PA13	I/O	GPIOA_13
13	PB21	I/O	GPIOB_21
14	PA18	I/O	GPIOA_18
15	GND	P	Ground
16	PA26	I/O	GPIOA_26
17	PB5	I/O	GPIOB_5 / UART_LOG_TXD
18	PA27	I/O	GPIOA_27
19	PA28	I/O	GPIOA_28 / FSDM
20	PA29	I/O	GPIOA_29 / FSDP
21	PA30	I/O	UART_RX / GPIOA_30
22	PA31	I/O	UART_TX / GPIOA_31

The signal type codes are used in the table:

- I:Input
- O:Output
- P:Power supply
- All I/O can be configured as UART/SPI/LEDC/I2S/I2C/PWM/IR/KEY/DMIC (A_30 is default UART RX, A_31 is default UART TX);

Strapping Pins:

Pin Name	Pin No.	Default State	Description
PB5	17	Pull-up	1: Normal Mode (default) 0: Flash Download Mode

Power Supply:

Parameter	Symbol	Min	Typ.	Max	Units
Power supply voltage	VCC	3.0	3.3	3.6	V

RF Characteristics

1. Wi-Fi Radio Standard

Wi-Fi Wireless Standard	Description
Wi-Fi frequency range	2412MHz ~ 2484MHz (2.4GHz ISM Band) 5180MHz ~ 5825MHz (5GHz)
Wi-Fi wireless standard	IEEE 802.11 a/b/g/n
Wi-Fi wireless standard Modulation	DSSS, DBPSK, DQPSK, CCK and OFDM (BPSK/QPSK/16-QAM/64-QAM)
Wi-Fi wireless data rate	802.11a: 6/9/12/18/24/36/48/54 Mbps 802.11b: 1/2/5.5/11 Mbps 802.11g: 6/9/12/18/24/36/48/54 Mbps 802.11n: HT20 MCS0-7, HT40 MCS0-7

2. Bluetooth Specification

Category	Feature	Description / Value
General	Bluetooth Standard	LE (1 Mbps, 2 Mbps)
	Host Interface	UART
	Frequency Band	2400 MHz ~ 2483.5 MHz

	Number of Channels	40 channels for BLE
	Modulation	GFSK
RF	BLE Output Power (Class 1)	8 dBm (± 3 dB)
Sensitivity	Sensitivity @ PER = 30.8% (LE 1 Mbps)	-95 dBm
	Sensitivity @ PER = 30.8% (LE 2 Mbps)	-90 dBm
	Maximum Input Level	GFSK: -20 dBm

Module dimension drawing:

- Module dimension: L x W : 16x24 (± 0.20) mm

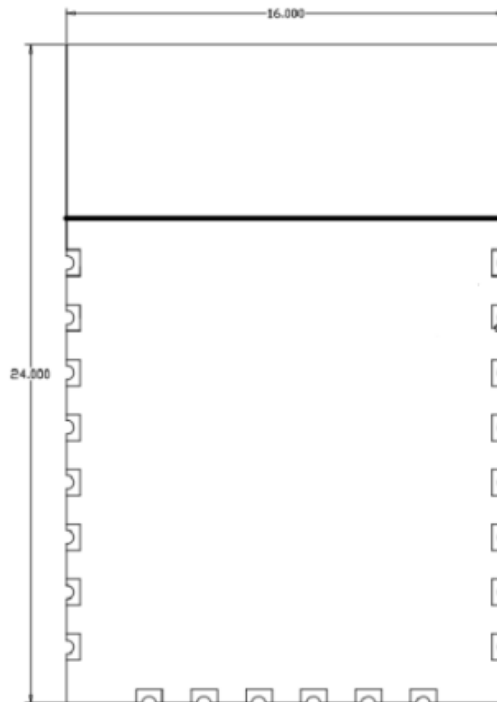


Fig 2 : Module dimension drawing

Recommendation Footprint Dimensions :

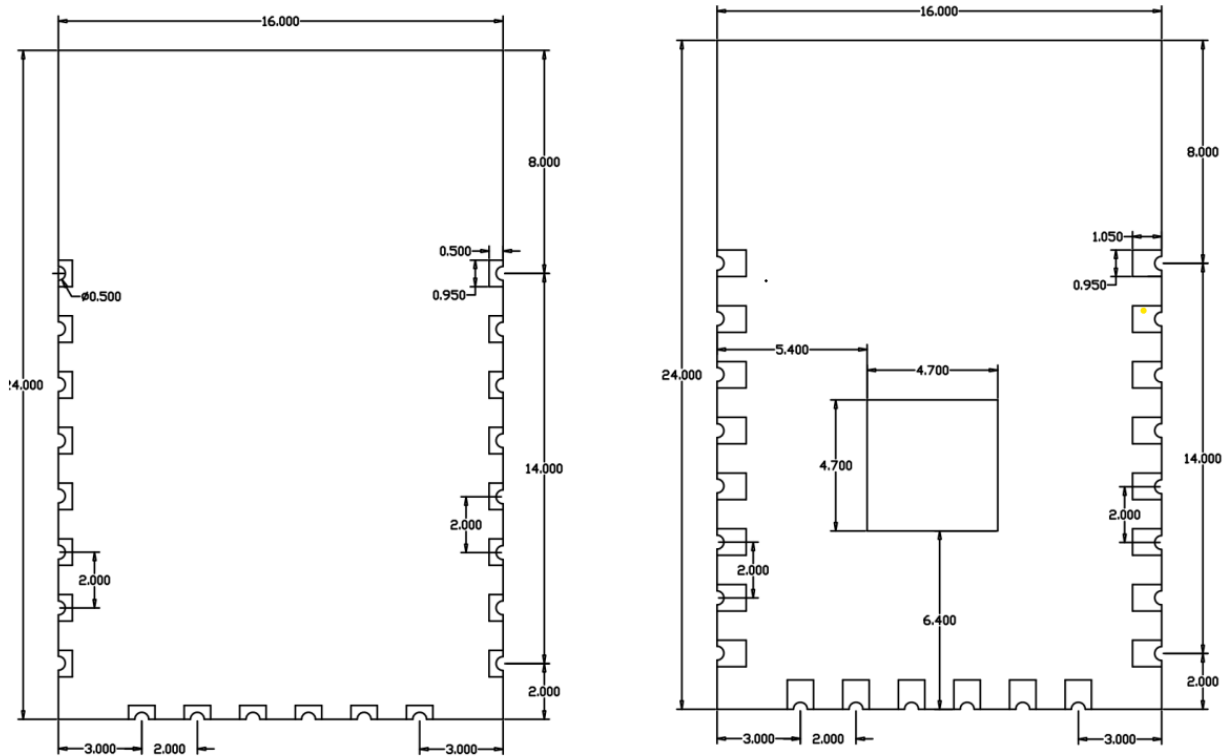


Fig 3 : Recommendation Footprint dimension

Product dimension:

- PCB Thickness : 0.8 ± 0.1 mm
- Product Height (including shield) : 2.6 ± 0.1 mm

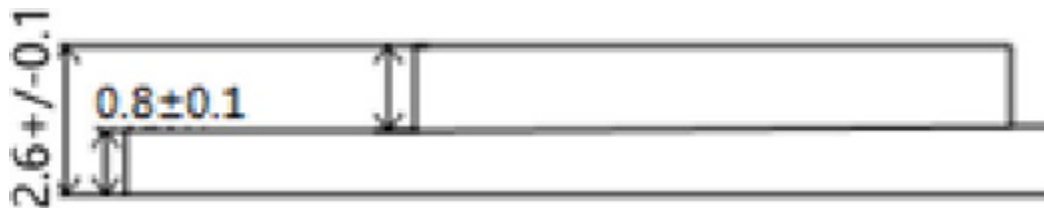


Fig 4 : Product Dimension

Module Electrical Characteristics:

1. Module Operating Conditions

Symbol	Parameter	Min.	Typ.	Max.	Units
VCC	Power supply voltage	3.0	3.3	3.6	V
Ta	Ambient operating temperature	-40	—	105	°C
Ts	Storage temperature	-65	—	150	°C

2. Module DC Characteristics

Symbol	Parameter	Conditions	Min.	Typ.	Max.	Units
VIH	Input-High Voltage	VIO = 3.3 V $\pm$ 10%	2.0	—	—	V
VIL	Input-Low Voltage	VIO = 3.3 V $\pm$ 10%	-0.3	—	0.8	V
VOH	Output-High Voltage	VIO = 3.3 V $\pm$ 10%	0.85 $\times$ VIO	—	—	V
VOL	Output-Low Voltage	VIO = 3.3 V $\pm$ 10%	—	—	0.15 $\times$ VIO	V

Module Block Diagram:

This module includes the chipset, crystal component, R/L/C components for RF matching, decoupling and RF radio antenna.

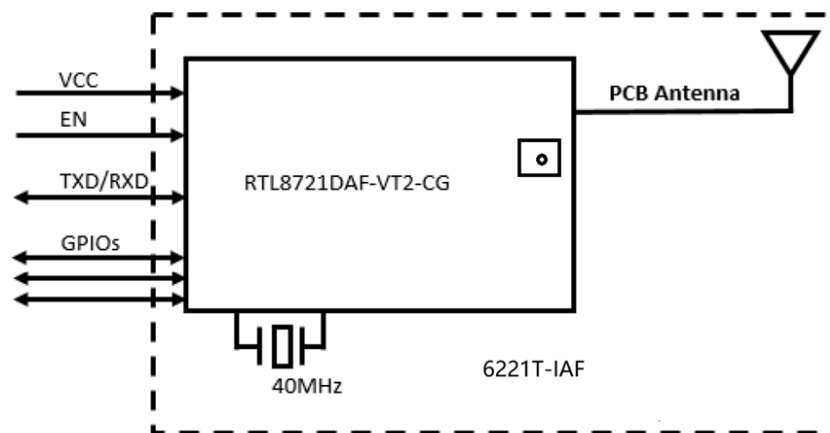


Fig 5 : Block Diagram

Recommended Reflow Profile :

Referred to IPC/JEDEC standard.

Peak Temperature: 260 °C

Number of Times: 2 times

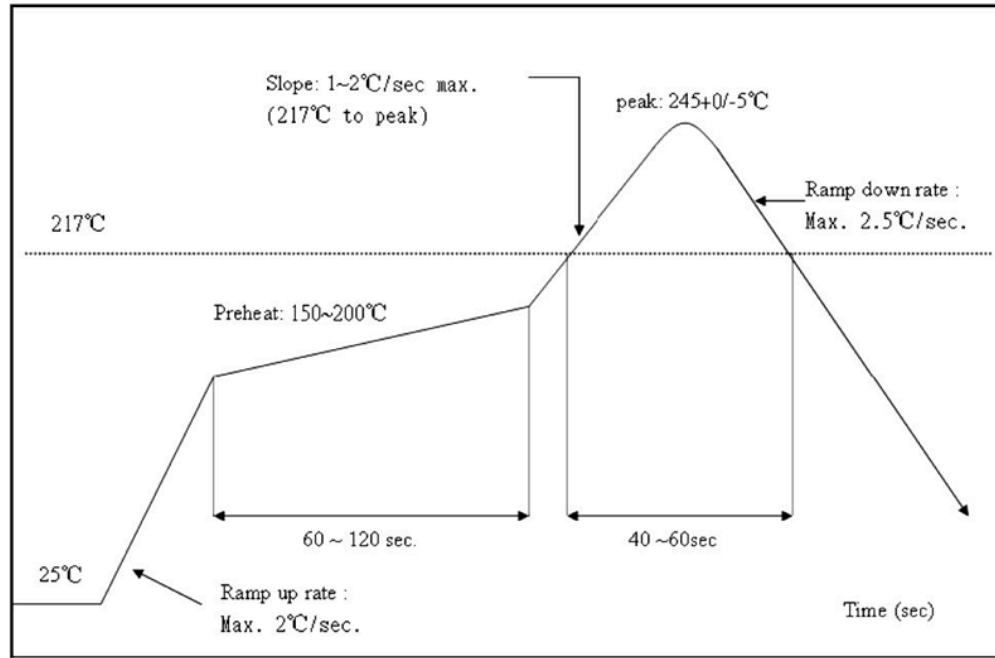


Fig 6 : Recommended Reflow Profile

Antenna clearance area requirements:

The installation positions on the motherboard are recommended in the following two ways:

Option 1: Place the module along the edge of the motherboard, and the antenna area extends beyond the edge of the motherboard.

Option 2: Place the module along the edge of the motherboard, and hollow out a section at the edge of the motherboard at the antenna position.

When applying the printed circuit board antenna to the Wi-Fi module, make sure that the distance between the printed circuit board on the main board and other metal devices is 16 millimeters. The shaded area in the figure needs to be kept away from metal devices, sensors, interference sources, and other materials that may interfere with the signal.

If there is no corresponding distance, please contact the FAE directly.

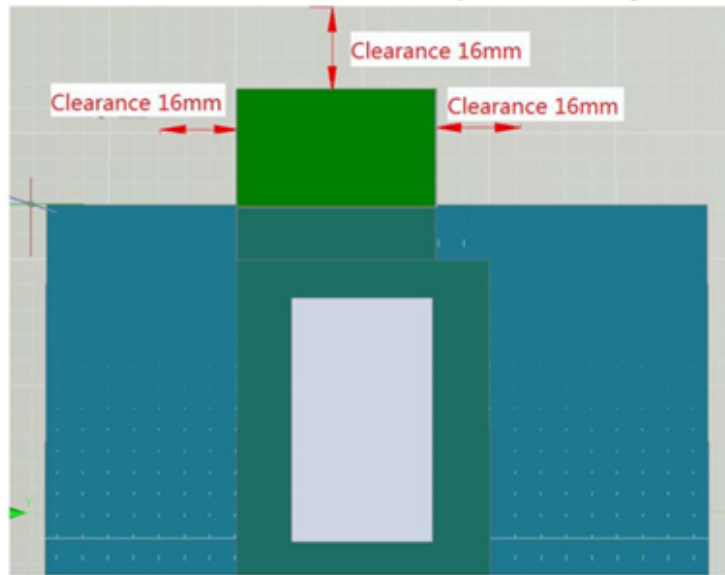


Fig 7(a): : Antenna clearance reference

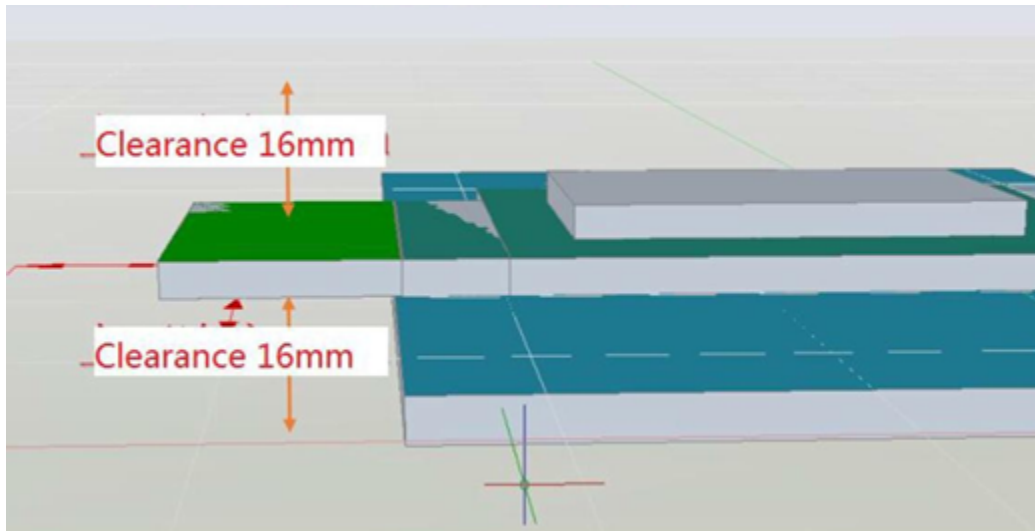
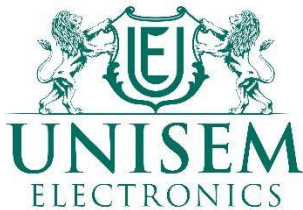


Fig 7(b) : Antenna clearance reference

For ordering USE_8721 please contact us on info@unisemindia.com.



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